Dear Editors:

We would like to submit the enclosed manuscript entitled **"Screening Under Signal Collapse: Choice and Architecture Experiments on Employer Decision Rules and Youth Access to Entry-Level Work"**, which we wish to be considered for publication in *Systems* for the Special Issue **"Navigating Digital Transformation: Leadership and Decision Making in Today's Systems."**

No conflict of interest exists in the submission of this manuscript, and the manuscript is approved by all authors for publication. I would like to declare on behalf of my co-authors that the work described was original research that has not been published previously and is not under consideration for publication elsewhere, in whole or in part. All authors listed have approved the manuscript that is enclosed.

This study asks how employers screen young applicants once generative artificial intelligence has made the written application cheap to produce, and whether the answer is a property of the people who screen or of the system they screen inside. Signalling theory predicts that a signal which everyone can produce at near-zero cost stops separating candidates, and platform evidence now shows exactly that collapse for written applications. What has been missing is the employer's side of the adjustment. We ran two linked experiments with 452 managers who make real shortlisting decisions in knowledge-intensive firms in China's Yangtze River Delta. Part A is a D-efficient discrete choice experiment: 6,328 choice tasks and 12,656 profile evaluations in which applicant profiles vary in AI-disclosure, process evidence, verified assessment results, institutional tier, referral, internship and salary, estimated by conditional logit, mixed logit with 500 scrambled Halton draws and a latent-class conditional logit. Part B holds the candidate pool fixed and randomises the screening interface itself—AI-score first, credential first, or verified-evidence first—so that the architecture is identified with preferences held constant by design, exactly as the preferences are identified with the architecture held constant in Part A.

The innovations of this paper are as follows.

(1) This study prices a signalling failure and shows that the repair is cheaper than the replacement. Expressing every attribute as a compensating salary differential puts the substitutes for the collapsed written signal in one metric. A verified assessment at the 90th percentile is worth 2.55 thousand CNY per month (95% CI 2.20 to 2.90), more than an elite degree at 1.83 and more than a referral at 1.73. Employers therefore prefer a costly, verifiable signal to an ascriptive one when it is available—but it almost never is, because virtually no applicant arrives holding a verified result. The binding constraint is the infrastructure, not employer demand, and that is a policy-actionable finding rather than an exhortation.

(2) This study shows that the penalty for disclosing AI assistance is large, conditional and largely repairable. Disclosure without corroboration costs an applicant 1.74 thousand CNY per month in compensating terms, about a sixth of monthly pay. Adding a portfolio of process evidence—drafts, revisions, prompts, supervision records—removes roughly three quarters of it, leaving −0.42, and removes all of it for the third of employers who already screen on evidence. This reframes the disclosure debate from whether declarations should be mandated, which invites an unenforceable answer, to what must accompany a declaration to make it informative. It also identifies a concealment trap: penalising honest disclosure without supplying a corroboration channel pushes applicants towards non-disclosure and destroys the information the rule was meant to create.

(3) This study shows that the equity–quality trade-off used to justify credential screening is an artefact of the interface, not a fact about candidates. Employer heterogeneity is discrete, not continuous: a three-class model recovers a credential-retreat regime (38.1%), a test-triage regime (28.4%) and an evidence-weighting regime (33.5%), and it beats the more flexible mixed logit out of sample (hit rate 0.642 versus 0.635; chance 0.333). The population-average disclosure coefficient of −0.741 describes no actual employer. Yet the architecture experiment dominates all of this: with the same managers and an identical candidate pool, evidence-first ordering raised the non-elite share of shortlists by 21.3 percentage points over credential-first and raised shortlist quality by 8.6 percentile points at the same time, at a cost of about 16 additional seconds per applicant. Screening on pedigree sacrifices quality as well as access, and the lever that fixes it sits with whoever configures the funnel.

We believe that this manuscript fits well within the scope of *Systems* and the Special Issue “Navigating Digital Transformation: Leadership and Decision Making in Today’s Systems”. Its object of study is a decision that digital transformation has quietly moved from people to configuration: what the first screen puts in front of a manager, and in what order. We show that this configuration choice moves hiring outcomes far more than the preferences of the individual managers do, which relocates a problem usually framed as individual bias into the domain of system design and leadership. The paper is quantitative and multi-method in the way the journal’s recent issues are—random-utility modelling of individual decision data, discrete heterogeneity, out-of-sample validation and a randomised manipulation of the decision environment—and it speaks directly to the Special Issue’s concern with how leaders make and govern decisions inside digitally transformed systems. It also complements the survey-based and macro-econometric contributions the Special Issue has attracted by supplying experimental identification at the level of the individual screening decision.

We deeply appreciate your consideration of our manuscript and look forward to receiving comments from the reviewers. Correspondence should be directed to Xinyu Cai at the following address:

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Thanks very much for your attention to our paper.

Very sincerely yours,

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